ECDLP leads, round 2 (2026-11-04)

Two frozen experiments have their own reports: ../anomalous-qtt-20261104 (NULL) and ../lowdeg-rank-20261104 (HOLDS). This report covers everything else from the round:

- three derivation checks;
- an adversarial review of a closed row;
- two literature amendments;
- new-lead synthesis, with the real novelty gate.

None of it is a measurement of attack cost. No disposition is changed. Proposals for the Coordinator are marked.

Independent validation of this round: VALIDATION-review.md (Sonnet validator; all verdicts confirmed, corrections applied).

Derivation checks (checks/, each runs in seconds to minutes)

check	claim tested	result
D1 d1_symnull.py	H-PSEUDO's recorded R ≈ 1.28–1.34 is the effect of a negation-closed factor base against an asymmetric null	**Supported.** Random negation-closed sets, with no curve structure, give R = 1.29 (N ≈ 2^17) and 1.31 (N ≈ 2^20) against the same null. The H-PSEUDO run code keys points by (x, y), so its factor base is ±-closed with signed logs. Matched to each rung's recorded parameters (the validator's recomputation), R against a negation-closed null is 1.01–1.03 ± 0.03 at the 2^20 rung (N_rep = 223,667, B = 512) and 0.96–1.02 across all rungs.
D2 d2_relations.py	low-degree rank deficits come from short multiplier relations Σc_i b_i ≡ 0 mod N	**Supported.** At p = 211 and 227, 88–90% of fresh random triples are exact. Every deficient triple has a relation with $\|c\| \le 2$, and both original p = 211 triples do ($b_2 \equiv -2b_3$ and $b_2 \equiv 2b_3$). Only two-term relations produce deficits; the 10 triples whose shortest relation has three terms are all exact. In the main low-degree run, two-term relations account for about 70% of deficient cells.
D3 d3_logfb.py	the discrete-log labelling's √p-scale QTT rank comes from the factor base's log set, not from the group law	**Supported in mechanism, two numeric predictions missed.** A log-AP factor base gives rank 12, 18 and 16 at p = 251, 509 and 1021, against 47–120 for small-x and random-log factor bases (E-block). The predicted ≤ 12 at every size and the ratio range [1, 4] were missed (observed 18 at L = 9, ratio 4.3–6.1).

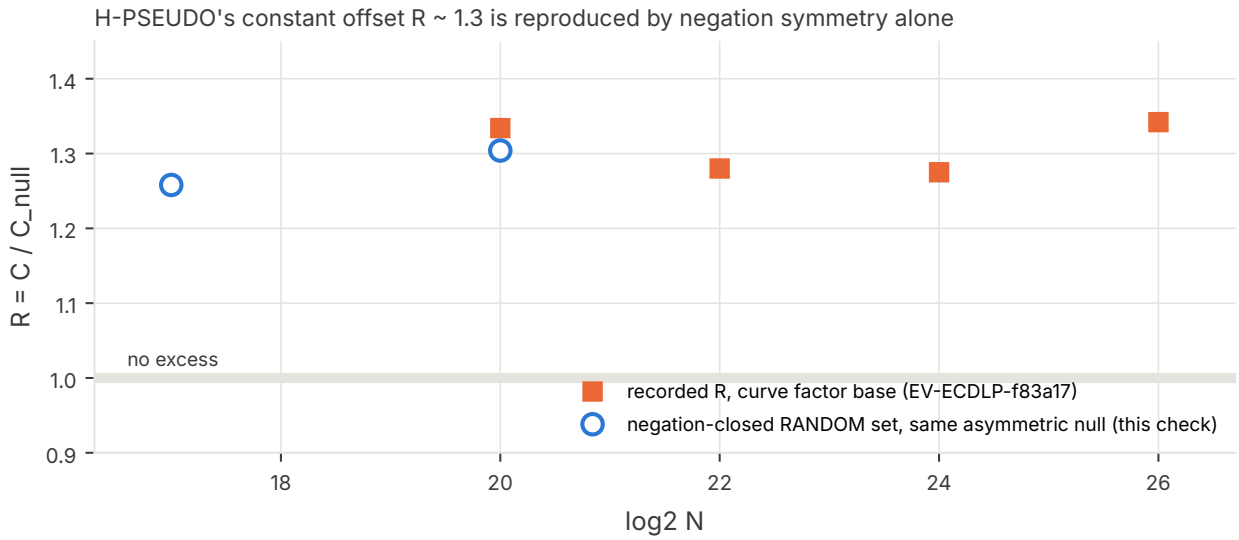

Figure 1: D1

These three checks went into rows as follows:

- **D1:** a note on the H-PSEUDO row, with no disposition change. Its revisit trigger now names a negation-matched null.
- **D2:** a note on the low-degree row.
- **D3:** a note on the holant row. The QTT branch reduces to the digit complexity of log(FB).

Proposed for the Coordinator:

- rerun the full H-PSEUDO ladder against a negation-closed null;
- consider filing the holant QTT branch closed_dominated by the character-sum row.

Review: LEAD-EXO-augmented-oracle-ggm (augmented-oracle-review.md; red-team, Opus)

The closure holds only for oracles computable from labels alone: endomorphism image as a label, label halving, incidence over a label factor base, and label-valued net. It fails for every coordinate-revealing oracle: x, y, full point, x-valued halving, field-valued jet and field-valued net. Two x-queries plus free local work give the logarithm. The row also states "jet O(1)", which CORR-20260727-003 withdrew ("No theorem exists", verified here). Its claim that the threshold oracle "needs private data" contradicts KN-FIND-982fdf, which says it is publicly computable (verified here). Its revisit_when has fired.

Proposed for the Coordinator: split the row. Keep it closed for label oracles, and open LEAD-EXO-coordinate-revealing-oracles, which states that the GGM gives no bound there and that the open question is index calculus itself. The row is **not edited** in this round.

Amendments from the 2026-11-01 scouting report (amendments-literature.md)

- **A1, applied as a scoped note to LEAD-ICE-generic-oracle-arity-cost-floor.** 3SUM-Indexing bounds (Golovnev et al., STOC 2020) assume preprocessing, an explicit list and |G| ≤ N^c. They give context only and do not extend the wall. 3SUM-Indexing is Goldstein et al. (WADS 2017), not the ESA 2016 paper.
- **A2, applied as a scoped note to LEAD-EXO-invertible-hint-subset.** Lange–Winterhof (COCOON 2002, AAECC 2003) and Kiltz–Winterhof (2004) are verified at abstract level only, because the theorems are behind publisher logins. The row's next step now begins by reading them.

New leads (new-leads-memo.md; synthesist, Opus) and the real novelty gate

The synthesist could not run lead_atlas.py, so its matches were emulated. Here the real gate (--match, several phrasings) was run:

proposal	top atlas hits (score)	decision
N1, weak curve classes as positive controls for the numeric instruments	SafeCurves audit (11), H-PSEUDO (10), kernel-of-reduction (10)	**filed** as LEAD-EXO-weak-class-instrument-calibration
N2, quadratic-phase (U^3) leakage of the small-x predicate	H-PSEUDO (12, linear phases only), invertible-hint (8)	**filed** as LEAD-EXO-quadratic-phase-small-x-leakage

N3, small-height rational (Farey) factor base	Coppersmith small-root (15), global-lift small height (12), cofactor ladder (12)	**not filed**: collides with existing rows

Items D1–D3 of the memo are the checks above. D4 predicts that the ML lead's planted homomorphic-leak control would train at chance. It is recorded here for that row's owner and was not tested in this round.

Not done

- Zhang–Liu (ePrint 2018/795) is still unread. eprint.iacr.org blocks this session, and no bypass was used.
- The Lange–Winterhof and Kiltz–Winterhof theorem statements are behind publisher logins.
- Conductor task T-5751 can only be closed from the owner's machine.